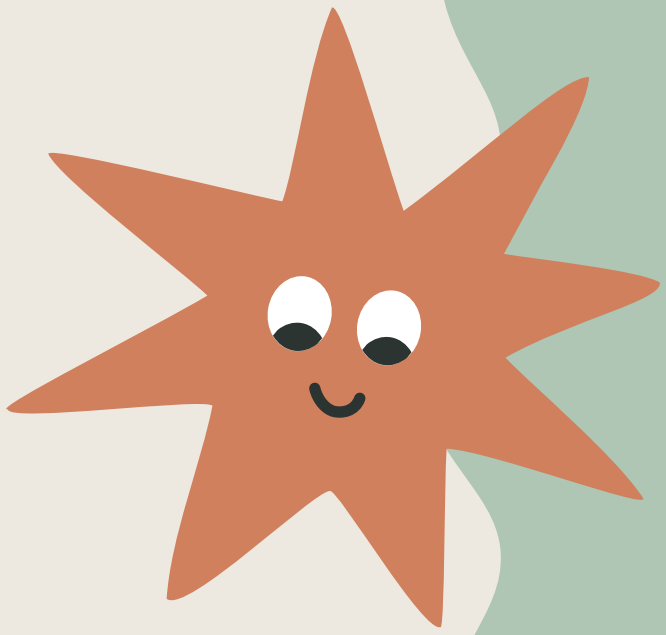
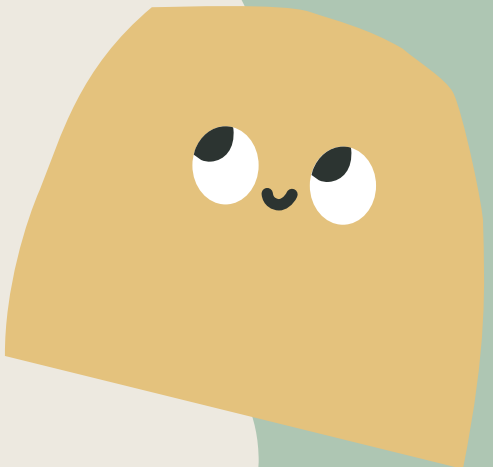
Predicate Logic in Artificial Intelligencee

Lecture 7





What Is Predicate Logic?

- **Predicate Logic** is a formal logic used to represent:
  - Objects, Properties and Relationships of Objects
  - General statements as well as Specific statements
- It extends propositional logic by introducing:
  1. Constants
  2. Variables
  3. Predicates
  4. Functions
  5. Quantifiers
- It is also called First-Order Logic (FOL)

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1. Constants

- A **constant** represents a specific object or individual
- Examples:
  - Ali, Ahmed, Sara, Lahore, Pakistan, Book1
- Student(Ali)
  - Here **Ali** is constant
- LivesIn(Ali, Lahore)
  - Here Ali = specific person
  - Lahore = specific place

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2. Variables

- A **variable** represents an unspecified or arbitrary object
- Common variable names include: x, y, z
- For example: Student(x)
  - Here x can represent any person
  - If **x = Ali** then **Student(x)** becomes **Student(Ali)**
- Variables become especially powerful when we use them with quantifiers

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3. Predicate

- A **predicate** represents a property or relationship
- For example: Student(Ali)
  - Means Ali is a student
- Here:
  - Student = predicate
  - Ali = argument
- Another example: Tall(Ali)
  - Means Ali is tall
  - Here **Tall** is predicate

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3.1. Unary Predicate

- A predicate with **one argument** is called a unary predicate
- Examples:
  - Student(Ali)
  - Tall(Ali)
  - Intelligent(Ali)
- General form
  - P(x)
  - Student(x) means x is a student

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3.2. Binary Predicate

- A predicate with **two arguments** represents a relationship between two objects
- For example:
  - Likes(Ali, Programming)
  - Means Ali likes programming
- Another example
  - Knows(Ali, Ahmed)
  - LivesIn(Ali, Lahore)
- General form
  - P(x, y)

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3.3. Higher-Arity Predicates

- Predicates can also have more than two arguments
- For example:
  - Teaches(Ali, AI, CS)
  - Means Ali teaches AI in Computer Science
- A predicate with three arguments is called a **ternary predicate**
- Predicates can represent relationships between multiple objects

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4. Functions

- A function maps one or more objects to another object
- For example:
  - FatherOf(Ali)
  - MotherOf(Ali)
  - CapitalOf(Pakistan)
- Functions are different from predicates
  - A predicate generally gives us **True / False**
  - A function returns an **object or value**

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5. Quantifiers

- Quantifiers are one of the most important concepts in Predicate Logic
- They allow us to express statements about:
  - All objects
  - Some objects
- There are two major quantifiers
  - Universal Quantifier (∀) For all
  - Existential Quantifier (∃) There exists

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5.1. Universal Quantifier

- The universal quantifier means, the statement is true for every object in the specified domain
- Example:
  - All humans are mortal
- We can represent this as:
  - ∀x Human(x) → Mortal(x)
- Read as:
  - For every x, if x is a human, then x is mortal
- Another example:
  - ∀x Student(x) → Learner(x)

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5.1.1 Understanding Universal Quantifier

- Suppose our domain contains:
  - Ali, Ahmed, Sara
- We have:
  - Student(Ali), Student(Ahmed), Student(Sara)
- And our rule is:
  - ∀x Student(x) → Learner(x)
- Then the system can infer:
  - Learner(Ali), Learner(Ahmed), Learner(Sara)
- Instead of writing separate rules for every student, we write one general rule
- This demonstrates the power of predicate logic

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5.2. Existential Quantifier

- The existential qualifier means, there is at least one object for which the statement is true
- Example:
  - Some students are intelligent
- We can represent this as:
  - ∃x Student(x) ∧ Intelligent(x)
- Read it as:
  - There exists at least one x such that x is a student and x is intelligent
- Another example:
  - ∃x Student(x) ∧ Knows(x, AI)

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Domain of Discourse

- The domain tells us what kinds of objects our variables can represent
- For example, our domain might be:
  - All people in a university
- Then Student(x) means:
  - x is a student in that university
- Another domain might be:
  - All animals
- Then Mammal(x) means:
  - x is a mammal
- The meaning of variables depends on the domain

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Example – Family Relationships

- Predicate logic is especially useful for relationships
- Suppose we've:
  - Parent(Ali, Ahmed)
  - Parent(Ahmed, Sara)
- We can create a rule:
  - $\forall x \forall y \forall z$ Parent(x, y) $\wedge$ Parent(y, z) $\rightarrow$ Grandparent(x, z)
- This means:
  - If x is the parent of y, and y is the parent of z, then x is the grandparent of z

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Multiple Quantifiers

- Predicate logic allows us to use more than one quantifier
- For example:
  - Every student has a teacher
- One possible representation is:
  - $\forall x \, \text{Student}(x) \rightarrow \exists y \, \text{Teacher}(y) \wedge \text{Teaches}(y, x)$
- Read it as:
  - For every student x, there exists a teacher y who teaches x
- This is much more expressive than propositional logic

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Free and Bound Variables

- Another important concept is the difference between **free and bound variables**
- Consider: Student(x)
  - Here x is a **free variable** because no quantifier controls it
- Consider: ∀x Student(x)
  - Here x is a **bound variable** because it is controlled by the universal quantifier
- Similarly: ∃x Student(x)
  - x is bound by the existential quantifier

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Advantages of Predicate Logic

- It can represent complex knowledge involving objects and relationships
- We can create general rules using variables
- We can represent concepts such as: All, Some, At Least One
- AI systems can derive new knowledge from existing facts and rules
- It is useful for representing structured knowledge in domains such as:
  - Expert systems
  - Natural language understanding
  - Planning
  - Reasoning systems

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Disadvantages of Predicate Logic

- It is harder to understand than propositional logic
- Reasoning can become computationally expensive as the knowledge base grows
- A human expert must provide or help construct the knowledge
- Cannot handle uncertainty effectively without extensions
- Logical representation of real-world problems can become complex
- Some problems in FOL are undecidable and lack guaranteed solutions

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Examples

- Write predicate logic for the following:
- All students are intelligent
- **$\forall x$ (Student(x) $\rightarrow$ Intelligent(x))**
- Some students are intelligent and hardworking
- **$\exists x$ (Student(x) $\wedge$ Intelligent(x) $\wedge$ Hardworking(x))**
- Every teacher teaches at least one student
- **$\forall x$ (Teacher(x) $\rightarrow$ $\exists y$ (Student(y) $\wedge$ Teaches(x, y)))**
- There is one teacher who teaches every student
- **$\exists x$ (Teacher(x) $\wedge$ $\forall y$ (Student(y) $\rightarrow$ Teaches(x, y)))**

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A Simple Method for Translating English ⟶ Predicate Logic

➢ **Every student has at least one teacher**

➢ Step 1 — Who are we talking about?

- Look for words like:
- Every, All, Some, At least one, There exists
- The word **Every** tells us that we need: ∀
- So we start with: ∀x

➢ Step 2 — What is x?

- The sentence says **"Every student"**
- So Student(x)
- Now we have: ∀x Student(x)

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A Simple Method for Translating English ⟶ Predicate Logic

➢Step 3 — What happens to each student?

- The phrase **“at least one teacher”** tells us that another object exists
- So introduce another variable: y
- and because at least one means **"there exists"**: ∃y
- Now we’ve: ∀x (Student(x) → ∃y …)

➢Step 4 — What is the relationship?

- The sentence says “A student **has** a teacher”
- For a more precise AI representation, we can define: Teacher(y)
- And Teaches(y, x)

➢So the final answer becomes:

- ∀x (Student(x) → ∃y (Teacher(y) ∧ Teaches(y, x)))

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